

Population Genomics of Polar Oceans Squid

Lucas Bastos^{1,3}, Charlotte Havermans^{2,3}, José C. Xavier^{1,4}, Catarina Silva¹



1. CENTRE FOR FUNCTIONAL ECOLOGY – SCIENCE FOR PEOPLE & THE PLANET (CFE), ASSOCIATE LABORATORY TERRA, UNIVERSITY OF COIMBRA CAMPUS AT FIGUEIRA DA FOZ, QUINTA DAS OLAIAS, 3080-183 FIGUEIRA DA FOZ, PORTUGAL
2. ALFRED WEGENER INSTITUTE HELMHOLTZ CENTRE FOR POLAR AND MARINE RESEARCH, BREMERHAVEN, GERMANY
3. UNIVERSITY OF BREMEN, BREMEN, GERMANY
4. BRITISH ANTARCTIC SURVEY, NATURAL ENVIRONMENT RESEARCH COUNCIL, CAMBRIDGE, UNITED KINGDOM

*Corresponding author: lucasbastos2000@hotmail.com

Introduction

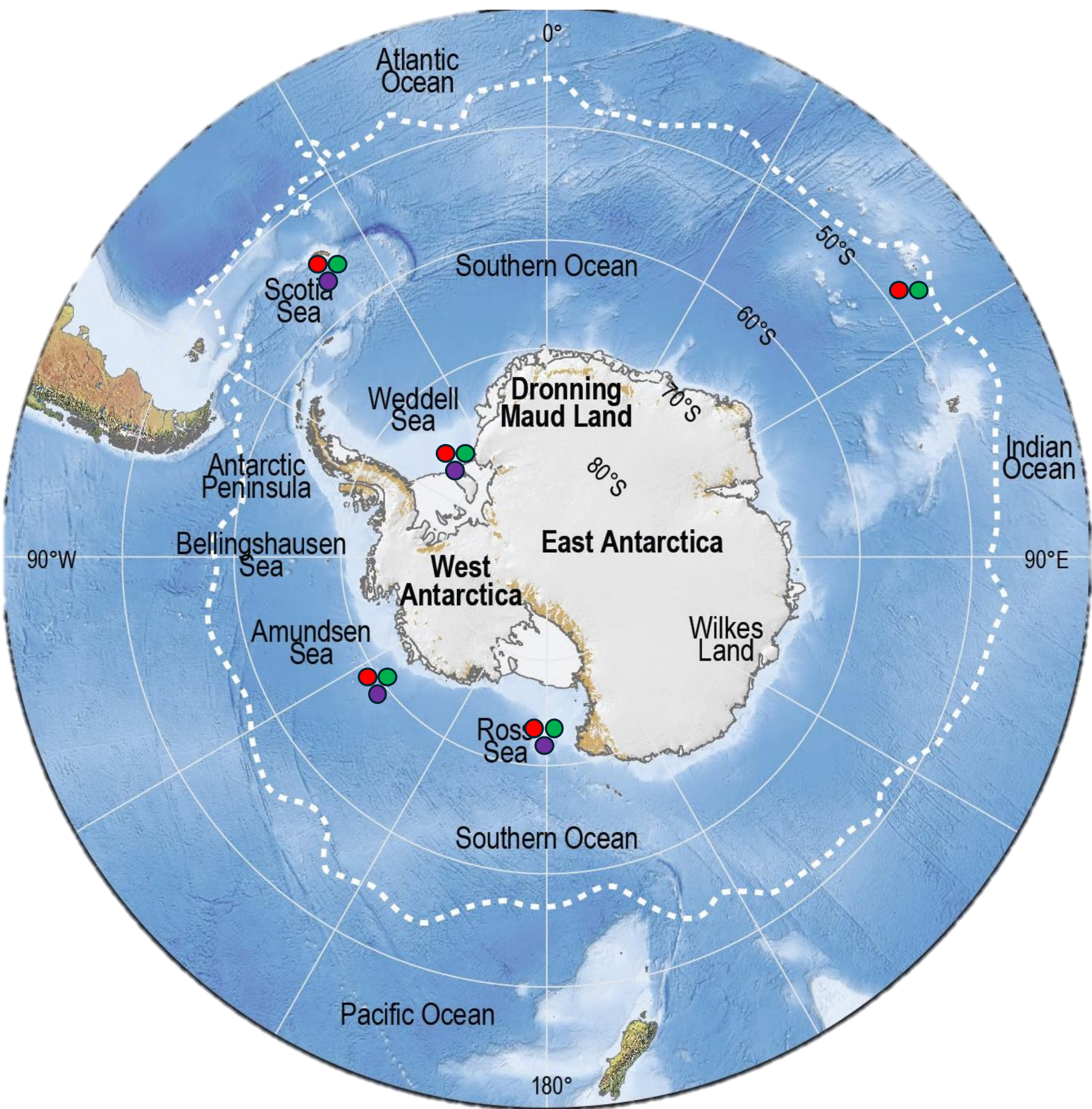
Polar regions are warming faster than the global average, driving rapid environmental changes that may profoundly affect marine biodiversity and ecosystem functioning¹. Squids play a key ecological role in both regions by linking lower and higher trophic levels^{1,2}. Understanding squid’s role in the ecosystem and their response to ongoing climate shifts is critical for predicting changes in marine food web dynamics and ecosystem resilience^{2,3}. Genomic markers such as SNPs allow the assessment of genetic diversity, population structure, connectivity, and local adaptation, providing insights on how populations may cope with environmental change^{3,4}. In this project we aim to **1) Unveil connectivity patterns between different species of Southern Ocean squids; 2) Assess genetic connectivity patterns of *Gonatus fabricii*; 3) Compare evolutionary and connectivity patterns between polar regions.**

Methodology

Existing unpublished sample collections from recent expeditions to both Southern and Arctic Ocean will be used alongside samples we aim to collect. When soft tissue is unavailable, cephalopod beaks will serve as alternative DNA sources, allowing inclusion of remote or under-sampled circumpolar regions². We aim to use ddRAD sequencing followed by Stacks pipeline to assemble loci *de novo* from the sequencing data for SNP calling⁵.

Antarctica

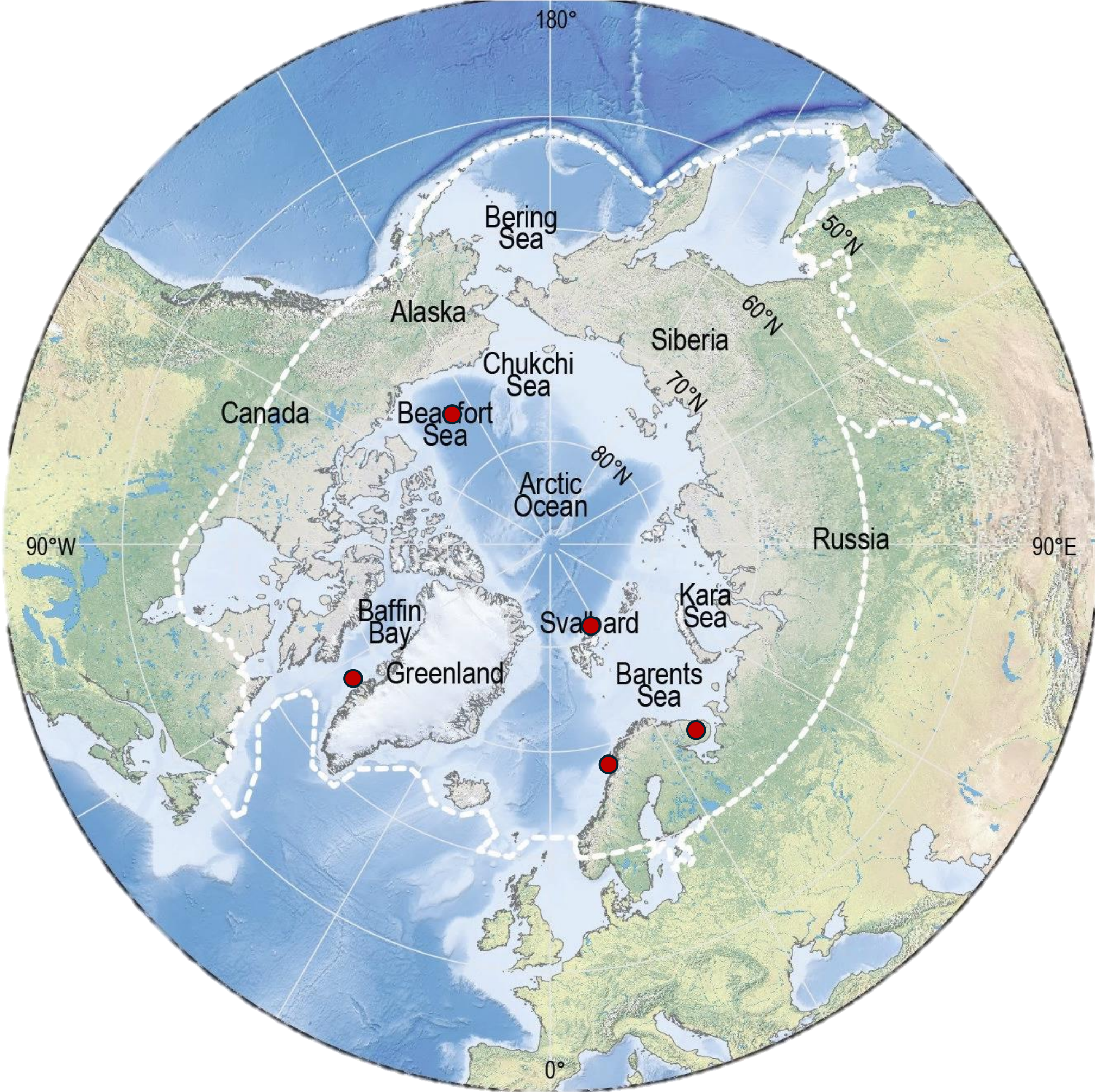
Figure 1. Map of the Southern Ocean showing sampling locations (points) used in this study.



- *Moroteuthopsis longimana*
- *Filippovia knipovitchi*,
- *Psychroteuthis glacialis*

Arctic

Figure 2. Map of Arctic Ocean showing sampling locations (points) used in this study.



- *Gonatus fabricii*

Expected Results

- We expect Arctic squid species to show greater population structure due to lower gene flow, but less pronounced and more recent local adaptation because of their recent evolutionary history.
- Antarctic squid populations, despite higher connectivity, facilitated by the Antarctic Circumpolar Current (ACC), may show stronger signs of local adaptation due to longer periods of evolutionary isolation.

References

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